Comparative Co-Occurrence Network Analysis of the New Testament Gospels and the Qur'an

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Abstract

This study seeks to analyze textual data from the four New Testament Gospels (Matthew, Mark, Luke, and John) and the Surahs of the Quran to construct and compare co-occurrence networks. The goal of this project is to evaluate how the distinct literary styles of chronological narrative and thematic discourse alter the structure of the resulting networks. Because the texts differ significantly in length and structure, the methodology requires data normalization, clustering coefficient comparisons, and degree distribution analysis to quantify structural similarities and differences.

1 Introduction

Practitioners of Abrahamic religions make up over half the world's population [1], and the holy texts of these faiths serve as the cornerstone for their respective beliefs and cultural traditions. Despite historical and contemporary contention between Christianity and Islam, the New Testament Gospels and the Qur'an share a deeply interconnected lineage stemming from Abraham. Shared figures bridge them conceptually, but the texts themselves are structured in fundamentally different ways. The four New Testament Gospels (Matthew, Mark, Luke, and John) are framed primarily as chronological narratives, whereas the Surahs of the Qur'an are formatted as a thematic discourse.

This study analyzes textual data from both the Bible and Qur'an to construct, visualize, and compare their underlying character networks. Specifically, this study evaluates how these distinct literary styles alter the resulting graph topology among shared entities. Because the texts differ significantly in length, scope, and structural organization, the methodology incorporates data normalization, clustering coefficient comparisons, and degree distribution analysis to quantify structural similarities and differences. Data for this study is drawn from the BibleData repository for the Gospels [1] and the Sahih International English translation via the Qur'an API [2].

This paper seeks to address three central research questions:

- Does a thematic text structure exhibit different network characteristics compared to a chronological narrative?
- How do the centrality metrics of shared characters shift between the two texts?
- Do both networks exhibit power-law degree distributions despite their fundamental structural differences?

Ultimately, this comparative network analysis serves to show that the shared principal figures rise above the structural settings of their respective texts. Beyond religious and literary divides, these texts represent a shared foundation of human hope and eternal faith.

2 Literature Review

In the noble pursuit of analyzing textual data, the application of network analytics provides an empirical perspective. Vague adjectives like *important* to describe characters are traded for betweenness centrality, *subjective groupings* yield to formal community detection algorithms, and prominence stops being a *matter of opinion* and starts being a function of degree density [3].

[1]According to the *World Population Review*, 32% of the global population identify as Christian, 25% Muslim and 0.2% Jewish

While traditional social networks require explicit interactions, character networks derived from literature often rely on co-occurrence [4]. Co-occurence[2] models emphasize relational proximity without requiring a line of spoken dialogue. This approach leverages the implicit, chronological nature of written narrative, capturing structural dependencies that direct dialogue models miss.

However, applying modern natural language processing (NLP) to ancient religious texts introduces distinct computational challenges. Standard Named Entity Recognition (NER) models struggle to identify entities labeled as `PERSON` when a character is referred to by multiple titles within the same text. Beyond entity resolution, establishing an appropriate threshold for what constitutes an implicit relationship remains a sticking point.

In biblical studies, network analytics serves a greater purpose than simply confirming that Jesus is the central network hub [5]. These methods can uncover distinct narrative communities [6], track how character influence shifts as a story progresses, and highlight bridges holding different scenes together. Unsurprisingly, a recurring theme across this literature is the difficulty of edge definitions. Whether defining by scene co-presence, verse co-occurrence, or a sliding window of N verses completely reshapes network density, degree distributions, and centrality rankings.

3 Methodology

To construct the co-occurrence networks, data was extracted from two separate repositories:

1) **Qur'an:** Textual data was gathered via the Qur'an API [2], utilizing the widely recognized Sahih International English translation. Translated by Emily Assami, Mary Kennedy, and Amatullah Bantley, this version is known for its modern, un-archaic language while maintaining a conservative interpretive. An English translation[3] was intentionally selected to facilitate standardized Named Entity Recognition (NER) and entity mapping.

2) **Gospels:** Character co-presence data for the New Testament was sourced from the open-source BibleData project [1]. Specifically, I utilized the pre-structured `PersonVerseApostolic.csv` file, which explicitly links figures within the same verse.

Shared Subgraph Sub-Set ($N = 18$)

There are **18 shared figures** out of the **198 Gospel nodes** and **38 Qur'anic nodes**:

- **Prophets and Kings:** David, Elisha, John, Jonah, Moses, Solomon
- **Divine and Angelic Figures:** Gabriel, Satan, YHVH (God)
- **Patriarchs:** Aaron, Abraham, Adam, Isaac, Jacob, Joseph, Lot, Noah
- **Matriarchs:** Mary

3.1 Global Topology

Nodes: characters

Edges: instances of co-occurence between characters

This brings up a difficult question to answer, how to define textual proximity for edges? - the bibledata defines relationships within verses, but the Qur'an is written in a different style requiring its own approach - also, the bibledata out of the box definition rersults in a pretty sparse network - Qur'an data preprocessing was set up to mirror that, co-presence established within each verse

either verse-level, sliding window, or surah-level. . .

[2] Social networks require explicit interactions between characters, co-presence characters are present in the same scene, and co-occurrence characters appear within a defined threshold

[3] Transliteration is converting from one writing system to another while preserving the pronunciation, translation is converting the meaning of a text from one language to another

Density: the proportion of actual connections in a network relative to all possible connections, indicating how interconnected the network is

Average Clustering Coefficient: quantifies the degree to which nodes in the graph tend to cluster together.

insert formula?

Average Path Length: the average number of steps along the shortest paths for all possible pairs of nodes. It measures the efficiency of information or mass transport in a network.

Diameter: the greatest distance between any pair of nodes, measured along the shortest paths connecting them.

3.2 Centrality Measurements

Degree Centrality Degree is a simple centrality measure that counts how many neighbors a node has. A node is important if it has many neighbors, or, in the directed case, if there are many other nodes that link to it, or if it links to many other nodes.

$$C_D(v) = \frac{\deg(v)}{N-1}$$

- $\deg(v)$: Number of edges connected to node v
- N: Total number of nodes in the network

Closeness Centrality In a connected graph, the normalized closeness centrality (or closeness) of a node is the average length of the shortest path between the node and all other nodes in the graph. Thus the more central a node is, the closer it is to all other nodes.

$$C'_B(v) = \frac{2}{(N-1)(N-2)} \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

- σ_{st}: Total number of shortest paths from node s to node t
- $\sigma_{st}(v)$: Number of those shortest paths that pass through node v

Betweeness Centrality Betweenness centrality quantifies the number of times a node acts as a bridge along the shortest path between two other nodes.

$$C_C(v) = \frac{N-1}{\sum_{u \neq v} d(v,u)}$$

- $d(v,u)$: Shortest path distance (geodesic distance) between node v and node u

Eigenvector Centrality Eigen-centrality is a prestige score which measures the node's influence. It is a nodal attribute, as is the case with all other centrality measures. Connections to high-scoring nodes contribute more to the score of the node in question than equal connections to low-scoring nodes. A high Eigen-centrality means the node is connected to many nodes with high scores.

$$\lambda x_v = \sum_{u \in N(v)} x_u = \sum_{u=1}^{N} A_{vu} x_u$$

3.3 Power-Law Distribution

A power law is a functional relationship between two quantities, where a relative change in one quantity results in a relative change in the other quantity proportional to a power of the change, independent of the initial size of those quantities

$$f(x) = \alpha x^{-k}$$

- In real networks, properties are often power-law distributed

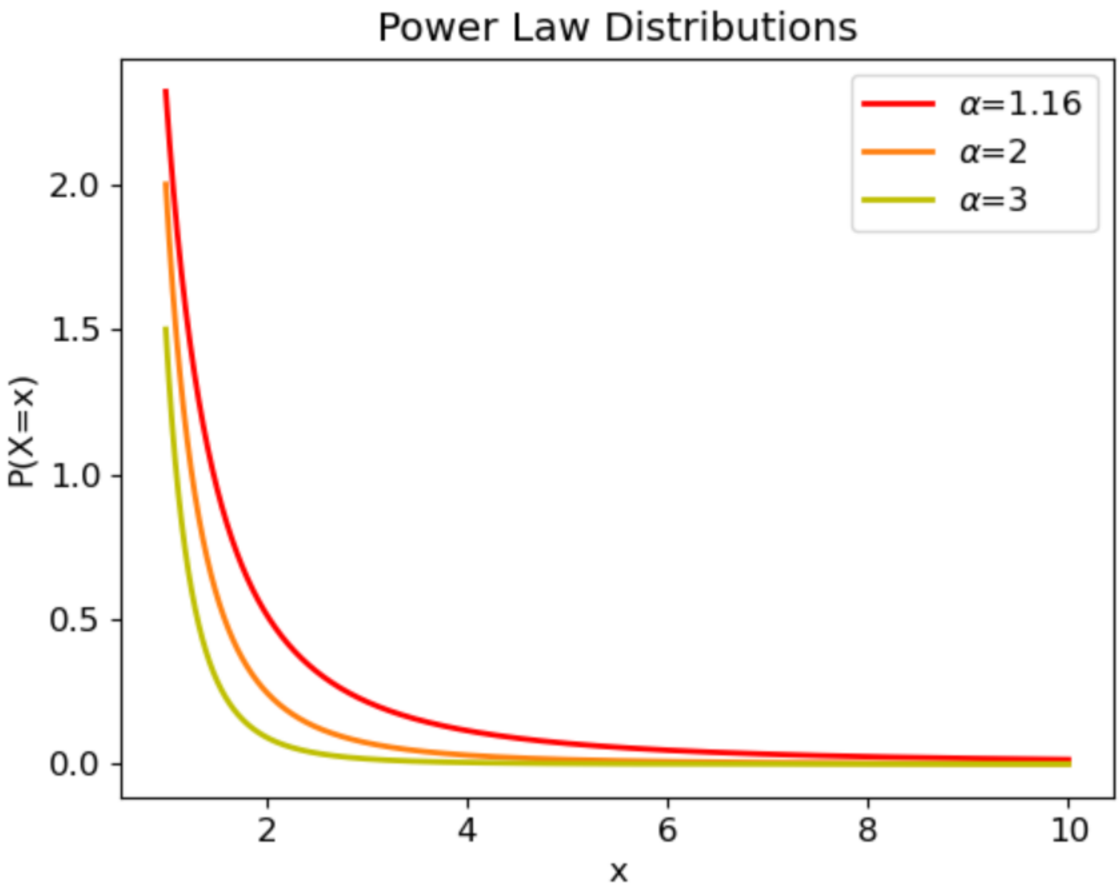

Figure 1: Sample Power Law Distribution Graph

3.4 Community Detection

Communities (i.e. modules or clusters): Sets of nodes with a relatively higher connection density within the cluster than between the clusters. High intra-cluster link density, lower inter-cluster link density

Community Detection: Identifies groups of nodes with dense connections within a graph. Primarily used in network analysis. Emphasizes the structure and relationships in the network. Examples: modularity optimization, label propagation.

Community Structure: Refers to the organization of nodes in a network into groups.

Louvain Method:

- For networks with weighted co-occurrence edges, the Louvain algorithm is the gold standard.
- Maximizes modularity score Q across the network.
- Efficient and effective for large networks.
- Uses a two-phase iterative optimization process. In phase 1, each node is moved to the community that yields the highest gain in modularity. In phase 2, the algorithm aggregates nodes of the same community and builds

a new network. These steps are repeated iteratively until modularity is maximized and cannot be increased further.

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

4 Results

Table 1: Overview of Textual Proximity Techniques

Network Name	Nodes ($\|V\|$)	Edges ($\|E\|$)	Density (D)	Avg. Clustering Coeff (C)	Avg. Path Length (L)	Diameter (d)
G $N = 1$	189	465	0.0262	0.7450	3.4567	8
G $N = 5$	198	2,657	0.1362	0.7871	1.9986	3
Q $N = 1$	37	106	0.1592	0.5825	2.1176	4
Q $N = 5$	37	184	0.2763	0.7786	1.7718	3
Q Surah-Level	37	326	0.4895	0.8441	1.5105	2

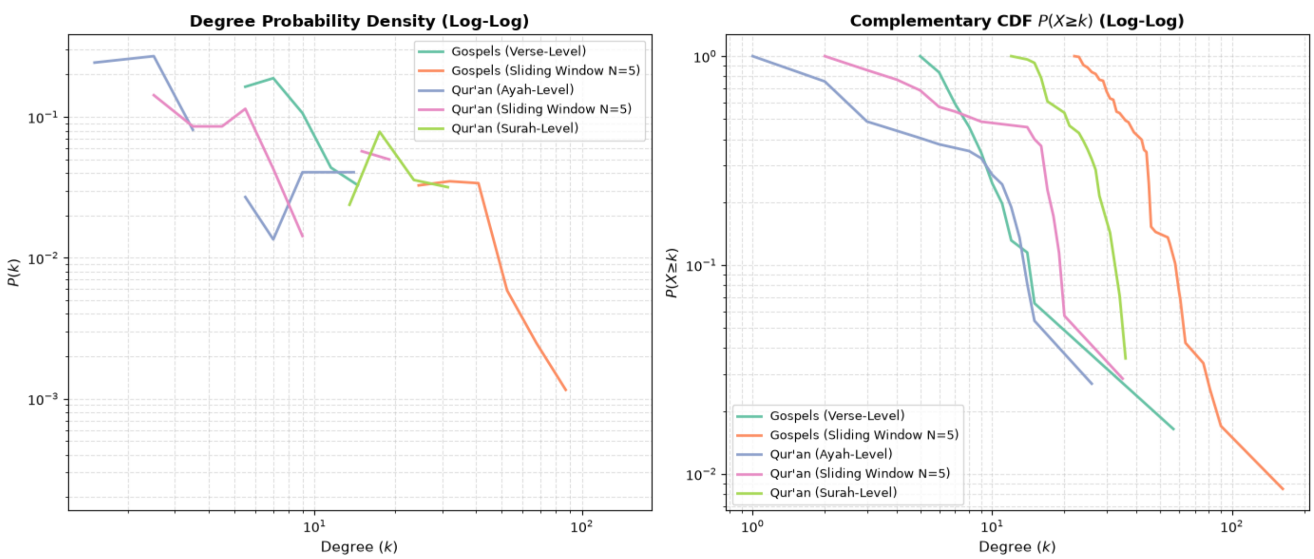

Figure 2: Power Law Distribution

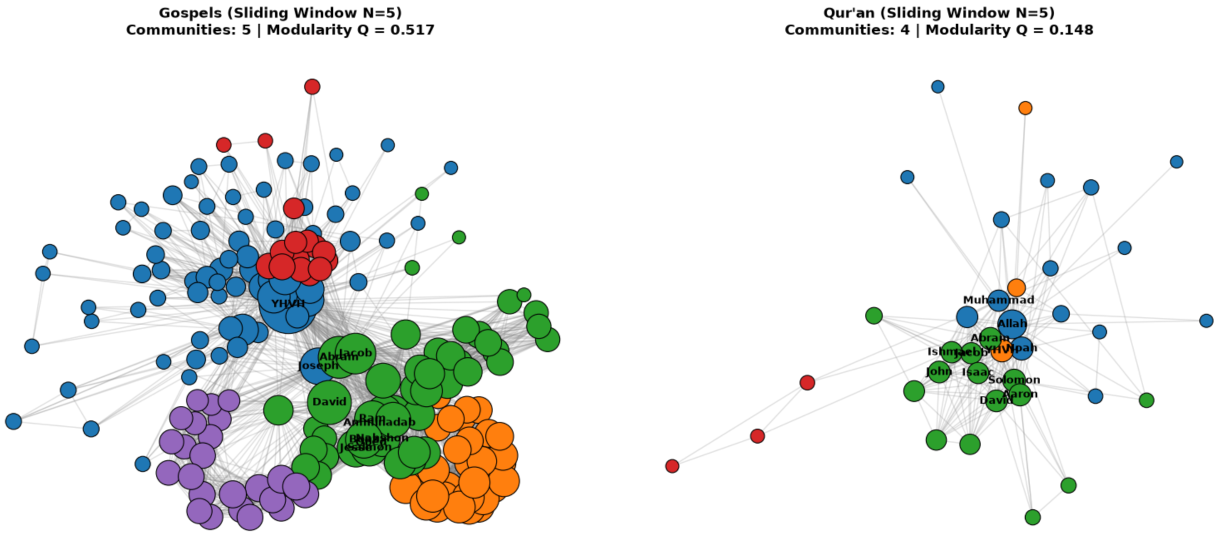

Figure 3: Community Detection

5 Discussion

6 Conclusion

7 References

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